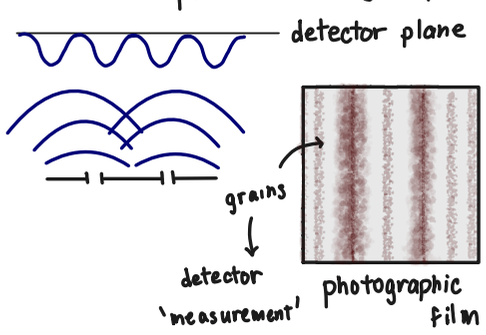
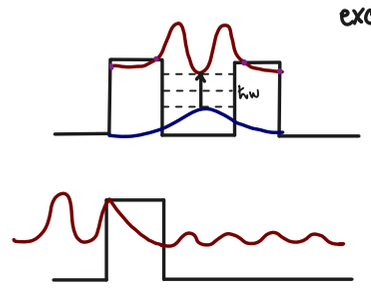
"Complex conjugates"

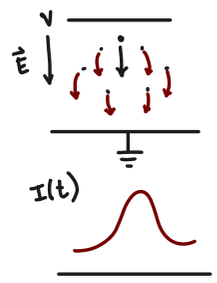
particle counting & probability



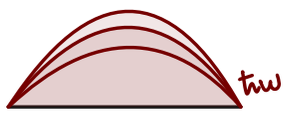
"macroscopic"
- typical currents ~ 10^{10} electrons
- typical mass 1g ~ 10^{23} atoms

"detector"
→ ovalanche mechanism
↳ matter wave excited w/ light
- escapes from bound state



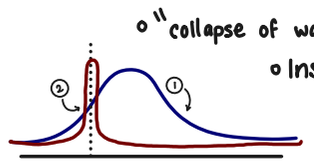


measurement



$E = n\hbar\omega_0$

- "collapse of wavefunction"
- Instantaneous change of ψ



* Born Rule: new empirical "law of physics" not derived from Schrödinger equation

probability of detecting @ particle @ $(\bar{x}, \bar{t})$

"Copenhagen Interpretation"

→ ψ is not 'real', only 'probability amplitude'

bosons – "amplitude", quantization

fermions – 0 or 1

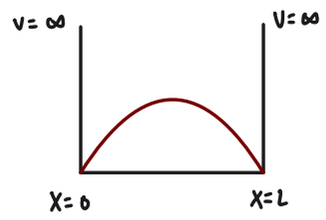
$\propto \boxed{|\psi(\bar{x},\bar{t})|^2}$

$= \psi^*(x,t)\psi(x,t)$

e.g. $\psi_0 e^{ikx} \cdot \psi_0 e^{-ikx}$

$\Rightarrow |\psi_0|^2$

Infinite Squarewell



$\psi(x) = A \sin kx$

$\lambda = \frac{2L}{N}$

- $\psi(x) = A\sin\frac{2\pi x}{\left(\frac{2L}{N}\right)} = A\sin\left(\frac{N\pi x}{L}\right)$

$\sqrt{\frac{2}{L}}\sin\left(\frac{N\pi x}{L}\right)$

- Find A by normalization

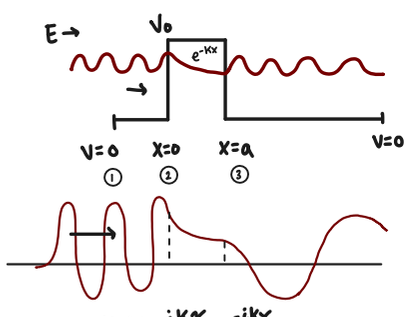
* total probability must equal 100% *

$\boxed{1} = \int dx\,|\psi(x)|^2 = \int_0^L dx\, A^2 \sin^2\left(\frac{\pi x}{L}\right) dx$

try: $\sin^2\theta = \frac{1-\cos 2\theta}{2}$: $\int ax\cos bx = \frac{1}{b}\sin bx$

$\rightarrow A^2\left(\frac{L}{2} - \frac{L}{\pi}\cancel{\sin\frac{\pi x}{L}}\Big|_0^L\right) \Rightarrow \boxed{1} = A^2\frac{L}{2} \rightarrow A = \sqrt{2/L}$

energy operator: $E_n = \frac{N^2\pi^2\hbar^2}{2mL^2}$



$\cos kx = \frac{e^{ikx} + e^{-ikx}}{2}$

region ①: Ae^{ikx}

region ②: Be^{-kx}

region ③: Ce^{ikx}

match @ boundary:

$x=0$: $e^{ikx} = e^{-kx}$, $x=0$

$\Rightarrow B = A$

$x=a$: $Ae^{-ka} = Ce^{ika}$

$\frac{C}{A} = \frac{e^{-ka}}{e^{ika}}$: $\left|\frac{C}{A}\right|^2 = |e^{-ka}|^2$ "complex number"

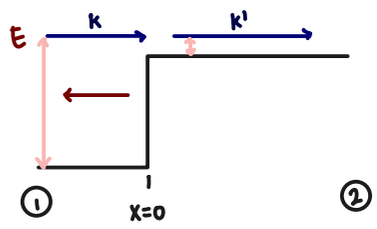
particle flux: $\frac{\#\text{ particles}}{\text{second}}$: $N \cdot v$ ← velocity

$v = \frac{p}{m} = \frac{-i\hbar\nabla}{m}$

$\psi^*\frac{p}{m}\psi = \psi^*\left(\frac{-i\hbar\nabla}{m}\psi\right)$ → $\frac{\hbar k}{m}$

$\propto \begin{pmatrix}\text{number}\\ \text{density}\end{pmatrix} \times$ velocity

unbound states "scattering"



region 1:

$\psi_1 = \cancel{A}^{1}e^{ikx} + Be^{-ikx}$ — R- Reflection coefficient ↳ $|r|^2 = r^*r$ is positive & Real

$\psi_2 = te^{ikx} + \left[\cancel{Ce^{-ikx}}\right]$ — 0 - input from left side only

2 unknowns

$k' = \frac{\sqrt{2m(E-V_0)}}{\hbar}$ $\quad E = \frac{\hbar^2 k^2}{2m}$

$\lambda = \frac{2\pi}{k}$, k is input

Boundary @ $x=0$

$\psi_1 = \cancel{e^{ik(0)}}^{1} + \cancel{re^{-ik(0)}}^{1} = te^{ik'(0)}$

$\Rightarrow \boxed{1+r=t}$

1st derivative

$\cancel{ike^{ik(0)}}^{1} - \cancel{ikre^{-ik(0)}}^{1} = ik'\cancel{te^{ik(0)}}^{1}$

↳ $ik(1-r) = ik't$

$\boxed{k(1-r) = k't}$

$r = t-1$

$k(1-(t-1)) = k't$

$k(2-t) - k't = 0$

$2k - (k+k')t = 0$

→ $\boxed{t = \frac{2k}{k+k'}}$

↳ $r = t-1 = \frac{2k' - (k+k')}{k+k'}$

↳ $\boxed{r = \frac{k-k'}{k+k'}}$

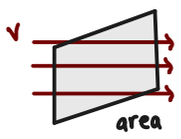
recall flux:

$\psi^*\left(\frac{-i\hbar\nabla}{m}\right)\psi = \frac{\hbar k}{m} = \frac{p}{m} = v$

$\boxed{|\psi|^2 v} = \frac{1}{v}\frac{\Delta x}{\Delta t}$

↑units?: normalization – $\int_L dx\,|\psi|^2 = 1$ ← unitless

↳ ψ has units $\frac{1}{\sqrt{L}}$



* in 3-D: $\frac{1}{\sqrt{v}} = \frac{1}{\sqrt{L^3}}$

- units of flux: $\frac{1}{\text{area}}\frac{1}{\Delta t}$ $\quad \frac{1}{L^3}\frac{\Delta x}{\Delta t}$ → units of length

flux in transmission: (need to square |t| to resolve probability wave)

$\frac{\hbar}{m}(|t|^2 k') = \frac{\hbar}{m}\frac{(4k^2k')}{(k+k')}$

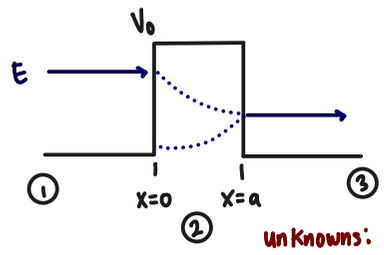
flux in reflection:

$\frac{\hbar}{m}(|r|^2 k) = \frac{\hbar}{m}\frac{(k-k')}{(k+k')}$

add these:

$|r|^2 k + |t|^2 k' = \frac{(k^2 - 2kk' + \overbrace{k'^2 + 4kk'}^{=+2kk'})k}{(k^2 + 2kk' + k'^2)} = (1)k$

↑ input flux



boundaries gives 4 conditions where $\psi = \psi^*$ @ boundary

① $(1)e^{ikx} + re^{ikx}$

② $Ae^{ikx} + Be^{ikx}$

③ te^{ikx}

$\kappa = \sqrt{\beta^2 - k^2}$

Kappa

unknowns:

@ $x=0$:

$e^{ik(0)} + re^{ik(0)} = Ae^{(0)} + Be^{(0)}$

↳ $1+r = A+B$

$ik(1) - rik(1) = A\kappa(1) - B\kappa(1)$

@ $x=a$:

$Ae^{\kappa a} + Be^{\kappa a} = te^{ika}$

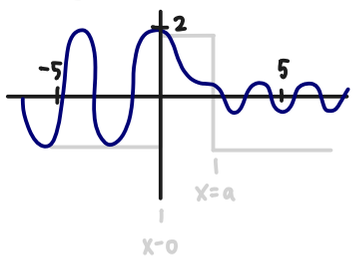
↳ $\kappa Ae^{-\kappa a} - \kappa Be^{\kappa a} = ik + e^{ika}$

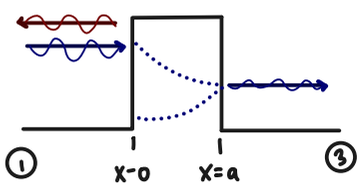
@ values:

A = 2

k = 3

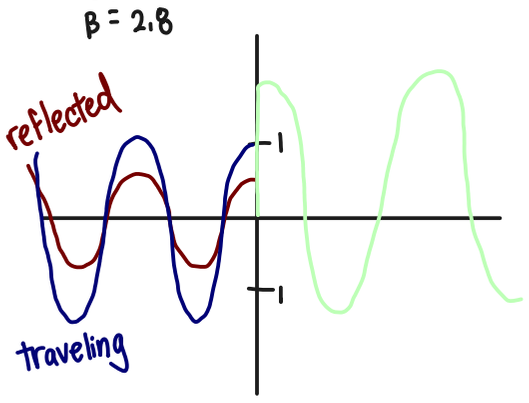
β = 10





@ k = 3

β = 2.8



$\cos(kx - \omega t)$ $\quad t=0$, $\omega t = \pi/2$

$\cos(kx)$

$\cos(kx - \frac{\pi}{2})$, $e^{i\theta} = \cos\theta + i\sin\theta$

classical waves: "impedance mismatch"